

Practical 8th

A system has RAM which can store four pages simultaneously to satisfy page requests. Write a program to calculate percentage page hit and page fault if the system uses Optimal page replacement technique for the page references entered by the user i.e. 1, 2,3,4, L,2,5, !,2,3,4, 5 (No of Frames=3)'

CODE :

```
#include<stdio.h>
int main()
{
    int frames=3;
    int pages[12]={1,2,3,4,1,2,5,1,2,3,4,5};
    int frame[3];
    int i,j,k,found;
    int pageFault=0,pageHit=0;
    for(i=0;i<frames;i++)
        frame[i]=-1;
    for(i=0;i<12;i++)
    {
        found=0;
        for(j=0;j<frames;j++)
        {
            if(frame[j]==pages[i])
            {
                pageHit++;
                found=1;
                break;
            }
        }
        if(found==0)
        {
            int pos=-1,farthest=i+1;
```

```

    for(j=0;j<frames;j++)
    {
        int k;
        for(k=i+1;k<12;k++)
        {
            if(frame[j]==pages[k])
                break;
        }
        if(k>farthest)
        {
            farthest=k;
            pos=j;
        }
    }
    if(pos==-1)
        pos=0;
    frame[pos]=pages[i];
    pageFault++;
}
}

printf("Page Hits = %d\n",pageHit);
printf("Page Faults = %d\n",pageFault);

float hitPercent=((float)pageHit/12)*100;
float faultPercent=((float)pageFault/12)*100;

printf("Hit Percentage = %.2f%%\n",hitPercent);
printf("Fault Percentage = %.2f%%\n",faultPercent);

return 0;
}

```

OUTPUT :



```
priti@LAPTOP-US9V80AL UCRT64 ~  
$ gcc --version  
gcc.exe (Rev13, Built by MSYS2 project) 15.2.0  
Copyright (C) 2025 Free Software Foundation, Inc.  
This is free software; see the source for copying conditions. There is NO  
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.  
  
priti@LAPTOP-US9V80AL UCRT64 ~  
$ notepad optimal.c  
  
priti@LAPTOP-US9V80AL UCRT64 ~  
$ gcc optimal.c -o optimal  
  
priti@LAPTOP-US9V80AL UCRT64 ~  
$ ./optimal  
Page Hits = 5  
Page Faults = 7  
Hit Percentage = 41.67%  
Fault Percentage = 58.33%  
  
priti@LAPTOP-US9V80AL UCRT64 ~  
$ |
```